

Continuous Distributions

Uniform Distribution

A continuous r.v X defined in the interval $[a, b]$ where a and b are finite, is said to follow a uniform distribution if the pdf of X is as follows:

$$f(x) = \frac{1}{b-a}, a \leq x \leq b \\ = 0, \text{ otherwise.}$$

X is said to be uniformly distributed. this also known as rectangular distribution.

The cdf of X is $F(x) = \int_a^x f(x)dx = \int_a^x \frac{1}{b-a}dx = \frac{x-a}{b-a}, a \leq x \leq b$

The mean and variance of uniform distribution is

$$\text{mean} = \mu_x = E(X) = \int_a^b xf(x)dx = \int_a^b \frac{x}{b-a}dx = \frac{b+a}{2}$$

$$E(X^2) = \int_a^b x^2 f(x)dx = \int_a^b \frac{x^2}{b-a}dx = \frac{b^2 + a^2 + ab}{3} \text{ hence variance is}$$

$$\sigma^2 = E(X^2) - \{E(X)\}^2 = \frac{(b-a)^2}{12}$$

The moment generating function is $M_X(t) = \int_a^b e^{tx} f(x)dx = \frac{e^{bt} - e^{at}}{t(b-a)}$

Using mgf the Mean =

$$E(X) = M'_x(t) \text{ at } t=0 \text{ similary variance is } \sigma^2 = E(X^2) - \{E(X)\}^2 = M''_x(t) - \{M'_x(t)\}^2 \text{ at } t=0$$

Examples:

1. Subway trains on a certain line run every half hour between mid-night and six in the morning. What is the probability that a man entering the station at a random time during this period will have to wait at least twenty minutes?

Ans. X is the r.v denotes the waiting time for the next train. X follows a uniform distribution

with pdf $f(x) = \frac{1}{30}, 0 \leq x \leq 30$. Therefore, $\text{prob}(X \geq 20) = \int_{20}^{30} f(x)dx = \int_{20}^{30} \frac{1}{30}dx = \frac{1}{3}$

2. If x is uniformly distributed with mean 1 and variance $4/3$ find $\text{Prob}(X < 0)$.

Ans. $\text{mean} = 1 = \frac{b+a}{2} = 1$ and $\sigma^2 = \frac{(b-a)^2}{12} = \frac{4}{3}$

Solving $a=-1$ and $b=3$. Therefore, $\text{prob}(X < 0) = \int_{-1}^0 f(x)dx = \frac{1}{4}$

3. Prove that for a rectangular distribution $f(x) = \frac{1}{2a}$, $-a \leq x \leq a$ find the mgf. hence find the mean and variance of X, Show that moments of even orders are given by $\frac{a^{2n}}{(2n+1)}$

In two dimension the pdf of uniform distribution is $f(x,y) = \frac{1}{\text{area of the bounded region}}$